ECE 310 Fall 2026
Lecture 5: Difference Equations and Block Diagrams

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Learning Objectives

- Interpret LTI systems defined by a Linear Constant-Coefficient Difference Equation (LCCDE).
- Identify whether a given LCCDE will yield a finite or infinite impulse response.



Linear Constant Coefficient Difference Equations

- *Linear Constant Coefficient Difference Equations* (LCCDE) are a popular representation of LTI systems.
- The system output $y[n]$ is a linear combination of system inputs and outputs.

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- K: number of output or *feedback* terms.
- M: number of input terms
- Couple of notes:
  1. Both K and M are finite while $M \geq 1$.
  2. We may express LCCDEs for non-causal systems as well.



Finite Impulse Response (FIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- If $K = 0$, i.e. no feedback terms, the system has a *finite-length impulse response* (FIR).
- In general, how can we find the impulse response of an FIR LTI system?



Infinite Impulse Response (IIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- If $K > 0$, the system has an *infinite-length impulse response* (IIR).
- In general, how can we find the impulse response of an IIR LTI system?



Infinite Impulse Response (IIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- If $K > 0$, the system has an *infinite-length impulse response* (IIR).
- In general, how can we find the impulse response of an IIR LTI system?
  - It's a bit more complicated.



Infinite Impulse Response (IIR) Systems

Example 1: Determine the impulse response $h[n]$ of the below system.

$$y[n] = \frac{1}{3}y[n-1] + x[n]$$



Infinite Impulse Response (IIR) Systems

Example 2: Determine the impulse response $h[n]$ of the below system.

$$y[n] = \frac{1}{3}y[n-1] + x[n] - 3x[n-1] + x[n-2]$$

